

POWER BI

Notes

Condensed from the Complete Training Course

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Module 1: Introduction to Power BI

What is Data Visualization?

- **Definition:** Converting raw numbers into charts, graphs, maps, and dashboards so people can understand information quickly and decide faster.
- Patterns and outliers that are invisible in a raw table jump out instantly in a chart.
- Can be static (printed/fixed) or interactive (click, filter, drill down) — Power BI builds interactive visuals.

Key Rule: Help the right people understand the right data at the right time — without reading hundreds of rows.

What is Power BI?

- **Definition:** A Microsoft business intelligence tool that connects to data, cleans it, builds interactive dashboards, and shares insights — no heavy coding needed.
- Workflow: Connect → Clean → Visualize → Share — done once, refreshes automatically.
- Integrates tightly with Excel, Azure, SQL Server, Teams, and SharePoint.

Example

Without Power BI: download CSV → clean in Excel → make charts → paste in PowerPoint → email weekly (repeat every week).

With Power BI: connect once to ERP → build dashboard → auto-refresh every Monday → team opens the live dashboard anytime.

Power BI vs Excel — Why Power BI?

- Handles millions of rows smoothly; Excel slows down and crashes.
- Visuals are interactive — clicking one chart filters all others automatically; Excel charts are static.
- Joining tables = drag-and-connect in Model View, instead of complex VLOOKUPS.
- Live/auto-refreshing data vs manual updates every time in Excel.
- Sharing = publish a link, everyone sees the same report; Excel sharing creates version confusion.

Key Benefits of Power BI

- Power BI Desktop is completely free; pay only for cloud sharing/collaboration.
- No-code drag-and-drop — build a working dashboard in under an hour.
- 100+ built-in data connectors (Excel, SQL, MySQL, Salesforce, Google Analytics, web APIs...).
- Scheduled automatic refresh (hourly/daily) with zero manual work.
- DAX formula engine for custom KPIs, running totals, and YoY comparisons.
- Native integration with the Microsoft ecosystem (Excel, Teams, SharePoint, Azure).

Power BI Ecosystem — Products

- Power BI Desktop — free Windows app; everything is built here (reports, model, DAX). Saves as .pbix.
- Power BI Service (app.powerbi.com) — cloud portal to publish, view, and collaborate on reports.
- Power BI Mobile — iOS/Android app for viewing dashboards on phones/tablets.
- Power BI Report Server — on-premises version for organisations that cannot use the cloud.
- Power BI Embedded — lets developers embed reports inside their own web apps.
- Power BI Premium — enterprise plan with larger limits and faster refresh (up to 48x/day).

Key Rule: Desktop = the design tool. Service = the publishing platform.

File Types You Should Know

Extension	What It Is
.pbix	Standard Power BI Desktop file — data, model, and report together

Module 2: Power BI Desktop — Interface Overview

The Three Views (Left Sidebar)

- **Report View** — The main canvas to build charts, tables, slicers, KPI cards. Has 4 areas: Canvas, Visualizations Pane, Fields Pane, Filters Pane.
- **Data View** — Shows loaded data in a spreadsheet-style grid; used to check data quality (correct date formats, numbers not text, no stray NULLs).
- **Model View** — Shows all tables as boxes connected by relationship lines — like a database schema; this is where relationships are created.

Example

Report View is like PowerPoint: drag a Column Chart onto the canvas, drop 'Month' on X-Axis and 'Sales Amount' on Y-Axis — the chart draws instantly.

Model View: drag CustomerID from the Sales table onto CustomerID in the Customers table — Power BI links them; filtering by customer name now filters sales too.

The Ribbon — Key Tabs

- **Home Tab** — Get Data, Transform Data, Refresh, Publish, and inserting New Visual/Text Box/Image/Buttons.
 - **Insert Tab** — New Page, Text Box, Image, Buttons, Shapes, and AI Visuals (Q&A, Smart Narrative, Decomposition Tree).
 - **Modeling Tab** — New Measure, New Column, New Table, Manage Relationships, Mark as Date Table, Row-Level Security (RLS).
 - **View Tab** — Themes, Performance Analyzer, Sync Slicers, Bookmarks Pane, Selection Pane.
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Module 3: Importing Data into Power BI

Data Types in Power Query / Data Model

Data Type	Example
Whole Number	1, 100, -5
Decimal Number	3.14, 99.99
Fixed Decimal (Currency)	₹1,250.50
Text	"Mumbai", "Product A"
Date / Time / Date-Time	01-Jan-2024, 10:30 AM
Duration	2 hours 15 minutes
True/False	TRUE / FALSE
Binary	Images, documents

Key Rule: Assigning the correct data type is critical — wrong types break DAX calculations, relationships, and visuals. Always check types after loading data.

Two Broad Categories of Data

- **Categorical Data** — Describes what something is — names, labels, groups (Nominal: no order, e.g. City; Ordinal: has order, e.g. Rating Low/Med/High).
- **Numerical Data** — Describes how much/how many — measurable quantities (Discrete: countable, e.g. Orders; Continuous: any value in a range, e.g. Revenue).

Common Data Sources

Category	Examples
File	Excel (.xlsx), CSV, JSON, XML, PDF, Folder
Database	SQL Server, MySQL, PostgreSQL, Oracle, MS Access
Power Platform	Power BI Datasets, Dataflows, Dataverse
Azure	Azure SQL, Blob Storage, Data Lake, Synapse Analytics
Online Services	SharePoint, Google Analytics, Salesforce, Dynamics 365
Other	Web URL, ODBC, Python Script, R Script

Module 4: Data Modelling in Power BI

What is a Data Model?

- **Definition:** A blueprint that tells Power BI how all your tables connect to each other.
- Stored in memory using the VertiPaq engine, which compresses data heavily for fast queries.

Key Rule: If a relationship is wrong or missing, charts show incorrect numbers silently — no error message, just bad data.

Star Schema vs Snowflake Schema

- **Star Schema** — One large central Fact table surrounded by smaller Dimension tables — Power BI is optimised for this; fast and simple.
- **Snowflake Schema** — Dimension tables are further broken into sub-dimensions (e.g., Products → Sub-Category → Category) — reduces repetition but adds complexity.

Fact Table vs Dimension Table

Fact Table	Dimension Table
Measurable, transactional data — actual events	Descriptive context — who, what, when, where
Sales transactions, orders, clicks, expenses	Customers, Products, Dates, Stores
Very high row count (thousands to millions)	Low row count (hundreds to thousands)
Numeric columns + Foreign Keys	Descriptive columns (Name, Category, Region)

Example — Retail Star Schema

Fact_Sales (centre): SaleID, DateKey, CustomerKey, ProductKey, StoreKey, Quantity, Amount

Dim_Date, Dim_Customer, Dim_Product, Dim_Store radiate outward — forming the star shape.

Creating Relationships (Model View)

- Switch to Model View → drag the key column from Table A onto the matching column in Table B → Power BI draws the relationship line automatically.
- The "1" side is always the dimension table (unique values); the "*" side is always the fact table (repeating values).
- Linked columns must share the same data type (Text-to-Text, Number-to-Number) or the relationship breaks.

Cardinality Types

Type	Use It?
One-to-Many (1:*) — one dimension row, many fact rows	Most common — always aim for this
One-to-One (1:1) — each row matches exactly one row	Rare - usually means tables should be merged
Many-to-Many (*:*) — multiple rows match multiple rows	Avoid — causes double-counting

Cross-Filter Direction

- **Single** — Filters flow from the '1' side to the '*' side only (recommended default).
- **Both** — Filters flow both ways — can cause circular dependencies; use only when specifically needed.

Key Rule: Always start with Single direction. Bidirectional relationships are a common source of hard-to-debug errors.

Module 5: Power Query — Data Cleaning & Transformation

What is Power Query?

- **Definition:** The data-cleaning workspace inside Power BI where you fix, reshape, and prepare data before it loads into the model.
- Open it via Home → Transform Data. Every click generates M code behind the scenes (Applied Steps panel).

Common Cleaning Operations

- Remove duplicate rows, remove unwanted columns, change column data types, rename columns/headers.
- Filter rows (keep/remove based on conditions), trim/clean text, replace values or errors with nulls/blanks.
- Split a column (by delimiter or position) or merge multiple columns into one.

Pivot vs Unpivot

- **Unpivot** — Wide format → long format. Turns multiple date/category columns into rows — Power BI strongly prefers long-format ("tidy") data.

Example

Before (wide): Product | Jan | Feb | Mar → Laptop | 50000 | 60000 | 55000

After unpivot (long): Product | Month | Sales → Laptop | Jan | 50000 (one row per product-month)

Adding New Columns

- **Custom Column** — Write an M formula, e.g. Revenue = [Quantity] * [Unit_Price].
- **Conditional Column** — No-code IF/ELSE builder, e.g. 'Large' if Quantity > 100 else 'Small'.
- **Column From Examples** — Type the desired output for a few rows; Power BI detects the pattern and fills the rest.
- **Date/Time Columns** — Extract Year, Month Name, Day of Week, Quarter, etc. from a date column.
- **Index Column** — Adds a sequential row number — useful as a unique row identifier.

Calculated Column vs Custom Column

Calculated Column (DAX)	Custom Column (Power Query / M)
Created in the Data Model; written in DAX	Created in Power Query Editor; written in M
Stored in model, recalculates on data refresh	Transformation happens before data loads
Has access to row context + relationships	Best for cleaning, reshaping, combining columns

Merge vs Append Queries

- **Merge Queries** — Combines two tables on a matching column — like a SQL JOIN / Excel VLOOKUP. Types: Left Outer (most common), Inner, Full Outer.
- **Append Queries** — Stacks two or more tables vertically (same columns required) — like SQL UNION ALL. Use Get Data → Folder to automate this for many files.

M Language — Quick Note

- "Mashup Language" — the scripting language behind every Power Query transformation; you don't need to learn it to use Power Query effectively.
- View the full code anytime via Home → Advanced Editor. Every query starts with let and ends with in.

Module 6: DAX — Data Analysis Expressions

What is DAX?

- **Definition:** The formula language of Power BI, used for custom calculations, KPIs, and logic that respond dynamically to filters, slicers, and visuals.
- Feels like Excel formulas, but works on entire columns/tables, not single cells.
- Results recalculate automatically every time a slicer is clicked or a filter changes — unlike static Excel results.

Three Types of DAX Calculations

- **1. Measures** — Dynamic calculation, computed on the fly every time a visual renders; responds to all active filters. Used for KPIs and totals. Created via right-click table → New Measure.
- **2. Calculated Columns** — Computed once per row at data refresh, stored permanently in the table. Used for labels/categories (e.g., Revenue Category Low/High). Created via New Column.
- **3. Calculated Tables** — A brand-new table generated entirely by a DAX expression, stored in the model (e.g., a Calendar table via CALENDARAUTO()).

Key Rule: If it summarizes data → Measure. If it works row by row → Calculated Column. If it returns a full table → Calculated Table.

Measure vs Calculated Column

Measure	Calculated Column
Evaluated when the report renders	Evaluated at data refresh time
Responds to slicers/filters (dynamic)	Fixed value per row (no filter response)
Not stored — calculated on the fly	Stored in the model (uses memory)
Best for: totals, ratios, KPIs, YTD	Best for: row categories, flags, extracted values

DAX Function Categories

Category	Best Used For
Aggregation (SUM, AVERAGE, COUNT...)	Measures
Filter (CALCULATE, FILTER, ALL)	Measures
Time Intelligence (YTD, SAMEPERIODLASTYEAR)	Measures
Text (UPPER, LEFT, CONCATENATE)	Calculated Columns
Logical (IF, SWITCH)	Columns + Measures
Date & Time (YEAR, MONTH, DATEDIFF)	Calculated Columns

Aggregation Functions

- SUM, AVERAGE, MAX, MIN — single-value totals/averages from a column.
- COUNT, COUNTA, COUNTROWS, DISTINCTCOUNT, COUNTBLANK — counting rows or distinct/blank values.
- SUMX, AVERAGEX, MAXX, MINX, COUNTX — "X" versions iterate row-by-row with a custom expression (needed when the calculation per row is dynamic, e.g. Quantity × Price).

Logical Functions

- IF(condition, true-result, false-result) — basic decision logic; SWITCH(TRUE(), ...) for multiple conditions cleanly.
- AND / OR / NOT — combine conditions (used inside IF).
- IFERROR, ISBLANK, ISNUMBER, ISTEXT — error handling and type checks.

Text Functions

- UPPER, LOWER, TRIM, LEN — case conversion, whitespace cleanup, text length.
- LEFT, RIGHT, MID — extract substrings by position.
- CONCATENATE, SUBSTITUTE, REPLACE — combine or swap text.

Date & Time Functions

- TODAY(), NOW() — current date/datetime.
- DATE, YEAR, MONTH, DAY, WEEKDAY — build or extract date parts.
- DATEDIFF, EOMONTH, DATEVALUE — date arithmetic and conversions.
- CALENDAR / CALENDARAUTO — generate a full Date table (returns a new table).

Filter Functions — The Power Behind DAX

- RELATED (pulls a value from a related table) and RELATEDTABLE (pulls a related table).
- VALUES, DISTINCT, SELECTEDVALUE, HASONEVALUE — work with the current filter context.
- ALL, ALLEXCEPT, ALLSELECTED, REMOVEFILTERS — remove or modify active filters.

CALCULATE — The Most Important DAX Function

CALCULATE lets you change the filters active during a calculation instead of just accepting the report's current filters. It underlies almost every advanced DAX measure.

```
North Sales = CALCULATE(SUM(Fact_Sales[Amount]), Dim_Region[Region] = "North")
All Time Sales = CALCULATE(SUM(Fact_Sales[Amount]), ALL(Dim_Date))
Sales Last Year = CALCULATE(SUM(Fact_Sales[Amount]), SAMEPERIODLASTYEAR(Dim_Date[Date]))
```

Example — Sales % of Total

North = ₹12L, South = ₹10L, East = ₹8L, West = ₹15L → Grand Total = ₹45L.

Numerator (plain SUM) follows the row's region filter; Denominator (CALCULATE + ALL) always uses the full dataset.

Result: North's % of Total = 12L / 45L = 26.7% — correctly ignoring the region filter only in the denominator.

Practical Measures Used in Real Projects

```
Total Revenue = SUM(Fact_Sales[Amount])
Gross Profit = [Total Revenue] - [Total Cost]
Profit Margin % = DIVIDE([Gross Profit], [Total Revenue], 0)
Sales YTD = TOTALYTD(SUM(Fact_Sales[Amount]), Dim_Date[Date])
Sales LY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))
YoY Growth % = DIVIDE([Total Revenue] - [Sales LY], [Sales LY], 0)
```

Key Rule: Use DIVIDE() instead of the '/' operator — DIVIDE safely returns 0 (or a chosen value) instead of an error when the denominator is 0.

Module 7: Visualizations — Charts & Visuals

Choosing the Right Chart

Goal	Best Visual
Compare categories	Bar / Column Chart
Show trend over time	Line Chart / Area Chart
Show part-to-whole	Pie / Donut Chart (few categories only) or Stacked Bar
Show a single KPI number	Card / Multi-row Card
Show relationship between 2 numeric values	Scatter Chart
Show geographic data	Map / Filled Map
Show hierarchy / breakdown	Treemap or Decomposition Tree
Detailed row-level data	Table / Matrix

Key Rule: Avoid pie charts beyond 4–5 categories — they become unreadable. Use a bar chart instead.

Core Chart Types

- Best for comparing categories; horizontal bars (Bar) work well for long category names, vertical (Column) for time-based comparisons.
- Best for trends over a continuous range like time; Area Chart adds shaded volume below the line.
- Use Matrix instead of Table when you need row/column grouping with subtotals (like a pivot table).
- Card shows one big number (e.g., Total Revenue); Multi-row Card shows several KPIs together.
- Filled Map shades regions by value (e.g., state-wise sales); Map plots points by location.

Slicers — Interactive Filters

- **Definition:** A visual filter control users click directly on the report — by category, date range, or numeric range.
- List, Dropdown, Between (numeric range slider), Relative Date (Last 7 days, This Month, etc.).
- Use Sync Slicers (View tab) to apply one slicer's selection across multiple report pages.

Formatting & Visual Pane Essentials

- Title, Background, Border, Data labels, Legend - found under the Format Visual (paint-roller icon) pane.
- Conditional formatting on table/matrix cells — background color or data bars based on value (e.g., red for losses, green for profit).
- Apply a Theme (View tab) for one-click consistent colors across the whole report.

AI-Powered Visuals

- **Q&A Visual** — Type a plain-English question (e.g., "total sales by region") and Power BI auto-generates the chart.
- **Smart Narrative** — Auto-generates a plain-text summary of what's shown in a chart (e.g., "Sales increased 12% this quarter").
- **Decomposition Tree** — Click into a total and explore which category/sub-category is driving it — drills down automatically.
- **Key Influencers** — Highlights the key factors driving an increase or decrease in a selected metric.

Module 8: Building Reports & Dashboards

Report vs Dashboard

Report	Dashboard
Built in Power BI Desktop, multi-page	Built in Power BI Service, single page only
Interactive — full filter/slicer/drill capability	Pinned tiles from one or more reports
The detailed, working analysis file	A summarized, at-a-glance view for executives
Source of truth — dashboards are built from it	Created by pinning visuals from reports

Page Design Best Practices

- Place the most important KPIs top-left (where the eye naturally goes first).
- Keep one clear purpose per page — don't cram every chart onto one screen.
- Use consistent colors and fonts across all pages (apply a saved Theme).
- Leave white space — a packed page is harder to read, not more informative.

Interactivity Features

- Clicking a category in one chart automatically filters/highlights all other visuals on the page (Power BI's default behaviour).
- Buttons + Bookmarks let you build custom navigation (e.g., a button that jumps to a 'Details' page).
- Tooltip pages — a small custom report page shown when hovering over a data point.
- Drillthrough — right-click a data point to jump to a detail page filtered to just that item.

Example

Create Bookmark 'Q1 View' with Quarter slicer = Q1 → assign a button 'Show Q1' → clicking it always resets the page to that exact filtered state.